

Module 11: Graph Neural Networks for Physical Dependency Discovery

Comprehensive Physics and Machine-Learning Note for RadiationEffects_proj2

Inference of directed physical-variable relations, coupling strengths, mechanism types, time lags, feedback groups, and physically admissible coupling sequences across Modules 2–5

Purpose of this revised note. Module 11 is restricted to one graph only: the *physical dependency graph*. Its nodes are physical variables, material coefficients, and external sources appearing in Modules 2–5. Its directed edges represent direct physical dependence: one quantity enters the constitutive law, source term, algebraic constraint, or evolution equation of another quantity. The goal is to determine whether a physics-constrained graph neural network can recover this dependency structure from synthetic trajectories and controlled interventions, quantify how edge strength changes with operating regime, identify feedback groups, and derive a physically admissible coupling sequence. Spatial mesh adjacency is not a Module 11 learning target, and numerical solver ordering is not optimized here. The finite-element solvers remain only as generators of synthetic field trajectories and physics residuals.

Meaning of “coupling sequence.” A physical dependency graph can contain directed cycles. Therefore, it does not always define one total order of variables. Module 11 derives a sequence by separating instantaneous algebraic relations from time-lagged dynamical relations and by condensing strongly connected feedback groups into single supernodes. The result is a sequence of physical influence groups, not a claim that nature executes a serial numerical algorithm.

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Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
AI	Artificial intelligence	Computational methods for prediction, inference, control, or decision making.
ML	Machine learning	Construction of a parametric model from examples and constraints.
NN	Neural network	Nonlinear function formed from trainable layers.
GNN	Graph neural network	Neural network operating on nodes and relations represented by a graph.
MPNN	Message-passing neural network	GNN in which edges transmit learned messages and nodes update hidden states from aggregated messages.
NRI	Neural relational inference	Latent-graph framework that jointly infers interactions and predicts dynamics.
GCN	Graph convolutional network	GNN based on normalized neighbor aggregation.
GAT	Graph attention network	GNN that learns data-dependent message weights.
DAG	Directed acyclic graph	Directed graph with no directed cycles.
SCC	Strongly connected component	Maximal set of nodes mutually reachable through directed paths.
PDE	Partial differential equation	Governing equation containing spatial or temporal derivatives.
ODE	Ordinary differential equation	Governing equation involving derivatives with respect to one independent variable.
DAE	Differential-algebraic equation	Coupled system containing evolution equations and algebraic constraints.
FEM	Finite element method	Numerical method used by Modules 2–6 to generate field trajectories; it is not the graph topology learned in Module 11.
POD	Proper orthogonal decomposition	Linear compression of field snapshots into dominant modes.

Symbol / acronym	Name	Meaning in this document
OOD	Out of distribution	Regime not represented by the training distribution.
UQ	Uncertainty quantification	Estimation of uncertainty in learned graph structure or predictions.
$\mathcal{G}_P = (\mathcal{V}_P, \mathcal{E}_P)$	Physical dependency graph	Directed graph whose nodes are physical quantities and whose edges denote direct physical dependence.
v_i	Physical-variable node	A field, coefficient, source, or state variable.
$i \rightarrow j$	Directed physical edge	Quantity i directly enters the law governing quantity j .
A_{ij}	Edge gate or strength	Learned nonnegative measure of the importance of edge $i \rightarrow j$.
z_{ij}	Edge type	Category such as algebraic, constitutive, dynamical, source, derivative, or reaction edge.
ℓ_{ij}	Edge lag	Physical time delay or discrete time index separating cause and response.
s_{ij}	Edge sign	Monotonic sign when meaningful: increasing, decreasing, nonmonotonic, or unknown.
$\mathbf{X}$	Node-feature matrix	Matrix of normalized physical descriptors for all variable nodes.
$\mathbf{e}_{ij}$	Edge features	Prior mechanism type, dimensional compatibility, candidate status, and intervention metadata.
$\mathbf{g}$	Global context	Radiation condition, bias, material identity, ambient temperature, and simulation time.
$\mathbf{h}_i^{(\ell)}$	Node embedding	Learned hidden state of node i at message-passing layer ℓ .
$\mathbf{m}_{ij}^{(\ell)}$	Message	Learned contribution sent from source node i to target node j .
C_j	Defect concentration	Concentration of defect species j .
C	Reduced defect field	Single effective defect concentration used in the current reduced executable model.
T	Temperature	Lattice or effective thermal field.
ϕ	Electrostatic potential	Solution of Poisson's equation.
$\mathbf{E}$	Electric field	$\mathbf{E} = -\nabla\phi$.
ρ	Space-charge density	Charge source entering Poisson's equation.
n, p	Carrier concentrations	Electron and hole number densities.

Symbol / acronym	Name	Meaning in this document
J_n, J_p	Carrier currents	Electron and hole current-density fields.
J	Total current	$J = J_n + J_p$.
D_C, D_j	Defect diffusivity	Temperature- and state-dependent defect transport coefficient.
D_n, D_p	Carrier diffusivities	Electron and hole diffusion coefficients.
μ_n, μ_p	Carrier mobilities	Drift coefficients modified by temperature and defects.
τ_n, τ_p	Carrier lifetimes	Effective recombination and trapping time scales.
G	Carrier generation	Electron-hole pair source.
R	Carrier recombination	Net carrier-removal rate.
S_C, S_j	Defect source	Radiation-induced defect-production term.
Q_{rad}	Radiation heat source	Thermalized deposited-energy rate.
Q_J	Joule heat	Electrical heating, usually $J \cdot E$.
$\text{do}(x = x_0)$	Intervention	Controlled setting or removal of a variable or physical mechanism.
$\mathcal{L}_{\text{state}}$	State-prediction loss	Error between predicted and reference variable trajectories.
$\mathcal{L}_{\text{edge}}$	Edge-existence loss	Loss for predicting whether directed edges are present.
$\mathcal{L}_{\text{type}}$	Edge-type loss	Loss for classifying the physical mechanism of each edge.
$\mathcal{L}_{\text{lag}}$	Lag loss	Error in predicted physical delay.
$\mathcal{L}_{\text{physics}}$	Physics-residual loss	Penalty on governing-equation residuals.
$\mathcal{L}_{\text{cons}}$	Conservation loss	Penalty on charge, carrier, defect, or energy inconsistency.
$\mathcal{L}_{\text{sparse}}$	Sparsity loss	Penalty discouraging unsupported edges.
$\mathcal{L}_{\text{prior}}$	Prior-consistency loss	Soft penalty for violating trusted physical knowledge.
$\mathcal{L}_{\text{int}}$	Intervention loss	Penalty when predicted edge effects disagree with controlled mechanism ablations.

1 Module 11 in one sentence

Module 11 uses a physics-constrained GNN to infer the smallest directed graph that is sufficient to explain how the physical quantities in Modules 2–5 influence one another across operating regimes.

A successful graph should answer six questions for every candidate relation $i \rightarrow j$:

1. Does the direct edge exist after conditioning on intermediate variables?
2. What physical mechanism does the edge represent?
3. In which direction does influence flow?
4. Is the influence instantaneous, delayed, or distributed over several time scales?
5. How strong is the edge in the present physical regime?
6. With what uncertainty can the edge be asserted?

The central target is not a graphical picture chosen for convenience. It is an identifiable physical object: a directed, typed, possibly state-dependent interaction graph constrained by the governing equations and tested by interventions.

2 Scope: one graph and only one graph

2.1 The physical dependency graph

Define

$$\mathcal{G}_P = (\mathcal{V}_P, \mathcal{E}_P). \quad (2.1)$$

The node set contains physical quantities such as

$$\mathcal{V}_P = \{C_j, \rho, \phi, \mathbf{E}, n, p, \mathbf{J}_n, \mathbf{J}_p, T, D_j, \mu_n, \mu_p, D_n, D_p, \tau_n, \tau_p, G, R, Q_{\text{rad}}, Q_{\text{J}}, \dots\}. \quad (2.2)$$

A directed edge $i \rightarrow j$ is present when the governing relation for j depends directly on i , after conditioning on explicitly represented mediators.

For example,

$$C_j \rightarrow \rho \quad (2.3)$$

is direct because charged defects enter the space-charge formula. By contrast,

$$C_j \rightarrow \phi \quad (2.4)$$

is usually an indirect path when ρ is represented explicitly:

$$C_j \rightarrow \rho \rightarrow \phi. \quad (2.5)$$

The distinction between direct and indirect dependence is essential. An unconstrained attention map may assign weight to both paths, but a physical dependency graph should prefer the mechanistically explicit factorization.

2.2 What is deliberately excluded

Module 11 does not learn graph edges between geometric mesh points or finite elements. Spatial fields are still generated and observed, but they are compressed into features attached to physical-variable nodes. The graph topology remains a variable-to-variable topology.

Module 11 also does not rank numerical update permutations, relaxation factors, solver blocks, or computational schedules. Numerical solvers are used only to produce synthetic trajectories and evaluate governing-equation residuals. Their internal order is not a label and should not contaminate the learned physical graph.

2.3 Why this narrower objective is scientifically stronger

The physical graph can be compared with equations, mechanism ablations, sensitivity derivatives, and controlled interventions. This gives a clear ground truth in synthetic experiments. In contrast, a numerical update order depends on implementation, hardware, tolerances, and preconditioning. Removing that second target avoids mixing physics discovery with software optimization.

3 What does “optimized physical dependency graph” mean?

The word *optimized* must be defined mathematically. The desired graph is not simply the densest graph that predicts well. A fully connected graph can fit many datasets but is difficult to interpret and can represent indirect or spurious associations. The target is a graph that balances prediction, physical consistency, intervention response, and parsimony:

$$\min_{\theta, \mathbf{A}} [\mathcal{L}_{\text{state}} + \lambda_e \mathcal{L}_{\text{edge}} + \lambda_t \mathcal{L}_{\text{type}} + \lambda_\ell \mathcal{L}_{\text{lag}} + \lambda_p \mathcal{L}_{\text{physics}} + \lambda_c \mathcal{L}_{\text{cons}} + \lambda_i \mathcal{L}_{\text{int}} + \lambda_s \mathcal{L}_{\text{sparse}} + \lambda_r \mathcal{L}_{\text{prior}}] . \quad (3.1)$$

Here $\mathbf{A}$ contains directed edge gates and θ contains neural-network parameters. An optimized graph should be:

- **predictively sufficient:** it predicts held-out trajectories and interventions;
- **physically consistent:** it respects equations, conservation laws, dimensions, and signs;
- **minimal:** unnecessary direct edges are penalized;
- **directionally correct:** source and target are not interchangeable;

- **regime aware:** edge strength can change with bias, damage, temperature, or radiation source;
- **uncertainty aware:** ambiguous edges are reported as uncertain rather than asserted confidently.

4 Physical variables and equations inherited from Modules 2–5

4.1 Defect evolution

For defect species j , a representative equation is

$$\frac{\partial C_j}{\partial t} = \nabla \cdot (D_j \nabla C_j - z_j \mu_j C_j \mathbf{E}) + S_j + R_j(\{C_k\}, n, p, T). \quad (4.1)$$

This equation implies candidate direct edges

$$T \rightarrow D_j, \quad (4.2)$$

$$D_j \rightarrow C_j, \quad (4.3)$$

$$\mathbf{E} \rightarrow C_j \quad \text{for charged-defect drift}, \quad (4.4)$$

$$S_j \rightarrow C_j, \quad (4.5)$$

$$C_k \rightarrow C_j \quad \text{through reactions}, \quad (4.6)$$

$$n, p \rightarrow C_j \quad \text{if carrier-assisted reactions are retained}, \quad (4.7)$$

$$T \rightarrow C_j \quad \text{through Arrhenius reaction rates}. \quad (4.8)$$

If a coefficient node such as D_j is represented explicitly, the path $T \rightarrow D_j \rightarrow C_j$ is preferable to a redundant direct edge $T \rightarrow C_j$, unless temperature also enters the reaction term independently.

4.2 Electrostatics

The space-charge density is

$$\rho = q \left[p - n + N_D^+ - N_A^- + \sum_j z_j C_j \right]. \quad (4.9)$$

Poisson's equation is

$$-\nabla \cdot (\epsilon \nabla \phi) = \rho, \quad (4.10)$$

and the electric field is

$$\mathbf{E} = -\nabla \phi. \quad (4.11)$$

These relations generate the direct chains

$$(C_j, n, p, N_D^+, N_A^-) \rightarrow \rho \rightarrow \phi \rightarrow \mathbf{E}. \quad (4.12)$$

The edge $\rho \rightarrow \phi$ is an algebraic constraint edge. The edge $\phi \rightarrow \mathbf{E}$ is a derivative edge. Their meanings differ from dynamic causal edges and should therefore have different edge types.

4.3 Carrier transport

Representative drift-diffusion currents are

$$\mathbf{J}_n = q\mu_n n \mathbf{E} + qD_n \nabla n, \quad (4.13)$$

$$\mathbf{J}_p = q\mu_p p \mathbf{E} - qD_p \nabla p. \quad (4.14)$$

Carrier continuity equations are

$$\frac{\partial n}{\partial t} = \frac{1}{q} \nabla \cdot \mathbf{J}_n + G - R, \quad (4.15)$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \nabla \cdot \mathbf{J}_p + G - R. \quad (4.16)$$

Constitutive coefficients may satisfy

$$\mu_n = \mu_n(C_j, T, \mathbf{E}, \dots), \quad \mu_p = \mu_p(C_j, T, \mathbf{E}, \dots), \quad (4.17)$$

$$D_n = D_n(\mu_n, T), \quad D_p = D_p(\mu_p, T), \quad (4.18)$$

$$\tau_n^{-1} = \tau_n^{-1}(C_j, T), \quad \tau_p^{-1} = \tau_p^{-1}(C_j, T). \quad (4.19)$$

The graph therefore contains state-to-coefficient edges, coefficient-to-current edges, and current-to-state edges. A model that merges all of these into one undifferentiated attention coefficient loses useful physics.

4.4 Thermal transport

A reduced thermal equation can be written as

$$\rho_m c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q_{\text{rad}} + Q_{\text{J}} + Q_{\text{rec}} + Q_{\text{def}}, \quad (4.20)$$

with

$$Q_{\text{J}} = \mathbf{J} \cdot \mathbf{E}. \quad (4.21)$$

Temperature then modifies defect diffusion, reaction kinetics, mobility, diffusivity, lifetimes, and possibly dielectric and thermal properties. This creates several feedback loops.

5 The candidate physical dependency graph for the current project

5.1 Reduced executable graph

The current reduced executable path supports the following direct dependencies:

$$T \rightarrow D_C, \quad (5.1)$$

$$D_C \rightarrow C, \quad (5.2)$$

$$(C, n, p, N_D^+, N_A^-) \rightarrow \rho, \quad (5.3)$$

$$\rho \rightarrow \phi, \quad (5.4)$$

$$\phi \rightarrow \mathbf{E}, \quad (5.5)$$

$$(C, T) \rightarrow (\mu_n, \mu_p), \quad (5.6)$$

$$(C, T) \rightarrow (D_n, D_p), \quad (5.7)$$

$$C \rightarrow (\tau_n^{-1}, \tau_p^{-1}), \quad (5.8)$$

$$(\mathbf{E}, \mu_n, \mu_p, D_n, D_p, \tau_n, \tau_p, G) \rightarrow (n, p), \quad (5.9)$$

$$(n, p, \mathbf{E}, \mu_n, \mu_p, D_n, D_p) \rightarrow \mathbf{J}, \quad (5.10)$$

$$(\mathbf{J}, \mathbf{E}, Q_{\text{rad}}) \rightarrow T. \quad (5.11)$$

The broader physics notes additionally permit charged-defect drift, multi-species reactions, carrier-assisted defect reactions, recombination heating, and defect-reaction heating.

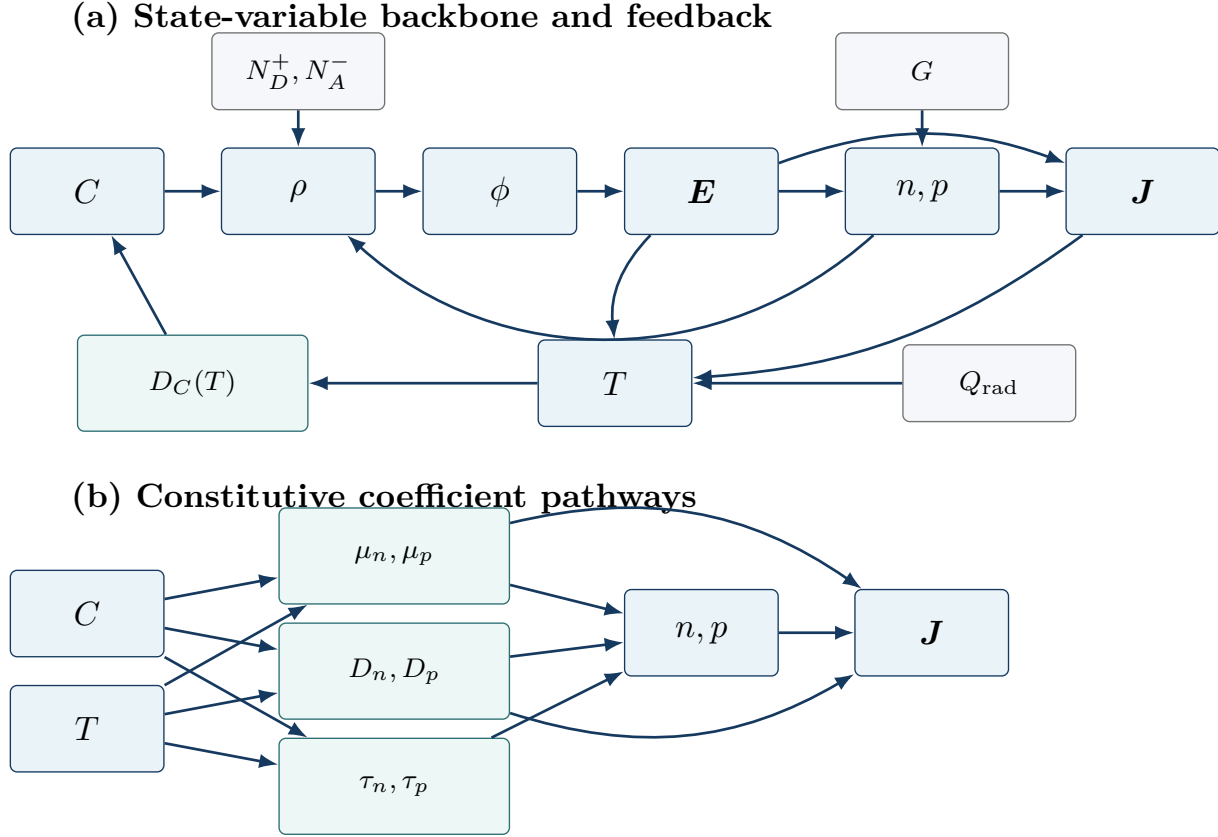


Figure 1: Candidate physical dependency graph for the reduced Module 6 physics. Panel (a) shows the state-variable backbone, major feedback loops, and external sources. Panel (b) isolates the constitutive pathways through mobility, diffusivity, and lifetime. Solid arrows are direct dependencies in the reduced model. Candidate mechanisms such as charged-defect drift are retained in the edge table and synthetic graph-family design rather than superimposed on this backbone diagram.

5.2 Direct, indirect, and redundant edges

Suppose a model predicts a strong edge $C \rightarrow \phi$. This association may be statistically real, but it is not necessarily a direct physical edge because C enters ρ , and ρ enters Poisson's equation. If both ρ and ϕ are nodes, the preferred representation is

$$C \rightarrow \rho \rightarrow \phi. \quad (5.12)$$

A direct shortcut $C \rightarrow \phi$ should be penalized unless it captures an omitted mechanism. This motivates conditional prediction tests and sparsity penalties.

5.3 Edge mechanism taxonomy

Each edge should carry a type label:

1. **algebraic composition**: variables combine into another variable, such as $(n, p, C_j) \rightarrow \rho$;
2. **constraint**: a source determines an algebraic field, such as $\rho \rightarrow \phi$;
3. **derivative**: one field is a derivative of another, such as $\phi \rightarrow \mathbf{E}$;
4. **constitutive**: a state changes a material coefficient, such as $T \rightarrow \mu_n$;
5. **transport driving**: a field or coefficient controls a flux, such as $\mathbf{E} \rightarrow \mathbf{J}_n$;
6. **dynamic source**: a source changes an evolving state, such as $G \rightarrow n$;
7. **reaction**: one population affects another through a reaction rate;
8. **heating**: electrical or reaction quantities enter the thermal equation;
9. **external forcing**: radiation, bias, or boundary conditions drive an internal variable.

Typed edges make the graph more interpretable and reduce the burden on the neural network.

6 Coupling sequences derived from a physical graph

6.1 Why a total ordering may not exist

The graph includes feedback loops such as

$$(n, p) \rightarrow \rho \rightarrow \phi \rightarrow \mathbf{E} \rightarrow (n, p), \quad (6.1)$$

and

$$T \rightarrow D_C \rightarrow C \rightarrow \mu_{n,p} \rightarrow \mathbf{J} \rightarrow T. \quad (6.2)$$

A directed cycle prevents a standard topological ordering. Forcing the graph into a DAG would erase real feedback or reverse an edge incorrectly.

6.2 Time unrolling

A physical feedback loop becomes acyclic when causes and responses are assigned to explicit time indices. A generic time-lagged edge is

$$X_i(t) \rightarrow X_j(t + \ell_{ij}\Delta t). \quad (6.3)$$

For example,

$$T(t) \rightarrow D_C(t), \quad (6.4)$$

$$D_C(t) \rightarrow C(t + \Delta t), \quad (6.5)$$

$$C(t + \Delta t) \rightarrow \rho(t + \Delta t), \quad (6.6)$$

$$\rho(t + \Delta t) \rightarrow \phi(t + \Delta t), \quad (6.7)$$

$$\phi(t + \Delta t) \rightarrow \mathbf{E}(t + \Delta t). \quad (6.8)$$

Algebraic relations may have zero lag, while transport and reaction edges generally have nonzero physical response times.

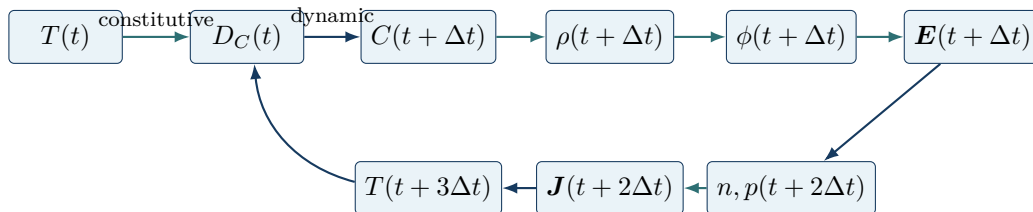


Figure 2: Illustrative time-unrolled physical dependency graph. Zero-lag algebraic or constitutive edges are shown separately from delayed dynamical edges. Time unrolling allows feedback without imposing an incorrect acyclicity assumption on the original physical system.

6.3 Strongly connected components

For a static directed graph, compute its strongly connected components. Every SCC is a maximal feedback group. Collapse each SCC into one supernode. The resulting condensation graph is always a DAG and can therefore be topologically ordered.

Let $\mathcal{C}_1, \dots, \mathcal{C}_K$ be the SCCs. The condensation graph is

$$\mathcal{G}_{\text{cond}} = (\{\mathcal{C}_k\}, \mathcal{E}_{\text{cond}}). \quad (6.9)$$

A physically admissible coupling sequence is then a topological ordering

$$\mathcal{C}_{\pi(1)} \prec \mathcal{C}_{\pi(2)} \prec \dots \prec \mathcal{C}_{\pi(K)}. \quad (6.10)$$

Variables within one SCC should be interpreted as a mutually coupled physical group rather than forcibly serialized.

6.4 What the GNN should output

The learned graph can therefore produce:

- a directed edge list;
- edge mechanism types;

- edge strength as a function of state;
- zero-lag versus delayed classification;
- physical lag estimates;
- SCC feedback groups;
- a topological order of the SCC condensation graph;
- uncertainty for every item above.

This is the precise sense in which Module 11 attempts to find a coupling sequence.

7 Graph fundamentals

7.1 Why graphs are the natural representation

A graph is a collection of entities and relations:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}). \quad (7.1)$$

The node set $\mathcal{V} = \{v_1, \dots, v_{N_v}\}$ contains physical quantities. A directed edge $(i, j) \in \mathcal{E}$ means that information flows from source node i to target node j . Each node has features $\mathbf{x}_i$, and each edge can have features $\mathbf{e}_{ij}$.

Graphs are appropriate because the Module 11 variables are heterogeneous and relational. Temperature, defect concentration, electric field, carrier density, mobility, and radiation source do not form a simple image-like array. Their meaning lies partly in how they influence one another.

7.2 Directed, undirected, homogeneous, and heterogeneous graphs

An undirected graph treats $i \leftrightarrow j$ symmetrically. That is generally unsuitable for physical dependency discovery because $\rho \rightarrow \phi$ is not equivalent to $\phi \rightarrow \rho$. The graph should be directed.

A homogeneous graph assigns the same semantic type to all nodes and edges. Module 11 is naturally heterogeneous: field nodes, coefficient nodes, source nodes, and reaction nodes have different meanings, while edge types distinguish constraints, constitutive laws, transport, reactions, and heating.

7.3 Adjacency matrices

For a directed weighted graph, define

$$A_{ij} = \begin{cases} a_{ij}, & i \rightarrow j \text{ is present,} \\ 0, & \text{otherwise.} \end{cases} \quad (7.2)$$

The convention is source row i , target column j . Because conventions differ across software, every implementation must document this choice and test it on a two-node graph.

A binary adjacency matrix expresses topology. A weighted adjacency matrix expresses effective influence. A typed graph can use one adjacency matrix for each mechanism class:

$$\mathbf{A} = \sum_{r=1}^R \mathbf{A}^{(r)}, \quad (7.3)$$

where r indexes relation types.

7.4 Permutation equivariance

Node ordering is arbitrary. If a permutation matrix $\mathbf{P}$ reorders node labels, the features and adjacency transform as

$$\mathbf{X}' = \mathbf{P}\mathbf{X}, \quad \mathbf{A}' = \mathbf{P}\mathbf{A}\mathbf{P}^T. \quad (7.4)$$

A node-level GNN should be permutation equivariant:

$$f(\mathbf{P}\mathbf{X}, \mathbf{P}\mathbf{A}\mathbf{P}^T) = \mathbf{P}f(\mathbf{X}, \mathbf{A}). \quad (7.5)$$

This property prevents the network from assigning meaning to an arbitrary storage order.

7.5 Receptive field and depth

After one message-passing layer, a node receives information from its one-hop parents. After L layers, information can propagate across paths of length up to L . The variable graph is small, so two to five message-passing layers are normally sufficient. Excessive depth can cause oversmoothing, in which node embeddings become indistinguishable.

8 Neural-network foundations needed for GNNs

8.1 A neuron and a multilayer perceptron

A dense neural layer computes

$$\mathbf{h}^{(\ell+1)} = \sigma \left(\mathbf{W}^{(\ell)} \mathbf{h}^{(\ell)} + \mathbf{b}^{(\ell)} \right), \quad (8.1)$$

where σ is a nonlinear activation. A multilayer perceptron composes several such layers. GNNs use MLPs inside edge-message and node-update functions, but they also respect graph structure through aggregation.

8.2 Training as optimization

Given samples $\{(x_s, y_s)\}$, neural parameters are obtained by minimizing

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{s=1}^N \mathcal{L}(f_{\boldsymbol{\theta}}(x_s), y_s). \quad (8.2)$$

For Module 11, x_s is a physical-variable graph plus trajectory context. Targets can include next-state variables, edge existence, edge type, edge lag, intervention response, and graph-level uncertainty.

8.3 Normalization is physically essential

The physical variables span many orders of magnitude. Defect concentrations, carrier densities, electric fields, mobilities, and heat sources cannot be passed to one network in raw SI units without severe conditioning problems. A positive variable x can be transformed as

$$\tilde{x} = \frac{\log(x + x_{\text{floor}}) - \mu_{\log x}}{\sigma_{\log x}}. \quad (8.3)$$

Signed variables can use a signed logarithm,

$$\tilde{x} = \frac{\text{sign}(x) \log(1 + |x|/x_0) - \mu_x}{\sigma_x}. \quad (8.4)$$

Normalization statistics must be computed from the training split only.

8.4 Overfitting and leakage

Data leakage occurs if different time steps from the same synthetic case are divided across train and test sets. The network can memorize case-specific parameters. Entire physical cases, source realizations, and intervention pairs must remain in one split. Regularization, early stopping, dropout, weight decay, graph sparsity, and independent topology tests should all be used.

9 Message passing: the core GNN operation

9.1 General equations

At layer ℓ , a directed message from node i to node j is

$$\mathbf{m}_{ij}^{(\ell)} = \phi_m^{(\ell)} \left(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}, \mathbf{g} \right). \quad (9.1)$$

The target node aggregates incoming messages:

$$\bar{\mathbf{m}}_j^{(\ell)} = \text{AGG}_{i \in \mathcal{N}_{\text{in}}(j)} \mathbf{m}_{ij}^{(\ell)}. \quad (9.2)$$

The node then updates:

$$\mathbf{h}_j^{(\ell+1)} = \phi_v^{(\ell)} \left(\mathbf{h}_j^{(\ell)}, \bar{\mathbf{m}}_j^{(\ell)}, \mathbf{g} \right). \quad (9.3)$$

The aggregation must be permutation invariant. Sum aggregation is expressive and preserves accumulated influence; mean aggregation removes degree dependence; maximum aggregation selects dominant mechanisms; attention learns normalized weights.

9.2 Directed typed messages

For relation type r , use a relation-specific message function:

$$\mathbf{m}_{ij}^{(r)} = a_{ij}^{(r)} \phi_m^{(r)} (\mathbf{h}_i, \mathbf{h}_j, \mathbf{e}_{ij}, \mathbf{g}). \quad (9.4)$$

A constraint edge should not be forced to use the same parameters as a heating edge. Relation-specific functions encode this distinction.

9.3 Project-specific edge inference

A practical directed edge logit is

$$\boldsymbol{\alpha}_{ij} = \text{MLP}_{\text{edge}} \left[\mathbf{h}_i, \mathbf{h}_j, \mathbf{h}_i - \mathbf{h}_j, \mathbf{h}_i \odot \mathbf{h}_j, \mathbf{e}_{ij}^{\text{prior}}, \mathbf{g} \right]. \quad (9.5)$$

The output can parameterize the probabilities

$$p(z_{ij} = r) = \text{softmax}(\boldsymbol{\alpha}_{ij})_r, \quad (9.6)$$

where $r \in \{\text{none}, \text{constraint}, \text{constitutive}, \text{dynamic}, \text{reaction}, \text{heating}, \dots\}$.

Direction is handled by evaluating $i \rightarrow j$ and $j \rightarrow i$ separately. Symmetrizing the logits would destroy directionality.

9.4 Readout functions

Node readouts predict future physical descriptors:

$$\hat{\mathbf{z}}_j(t + \Delta t) = \text{Dec}_j \left(\mathbf{h}_j^{(L)} \right). \quad (9.7)$$

Edge readouts predict existence, type, strength, sign, and lag. Graph-level readouts can summarize uncertainty, number of SCCs, and whether the inferred graph is sufficiently supported by the data.

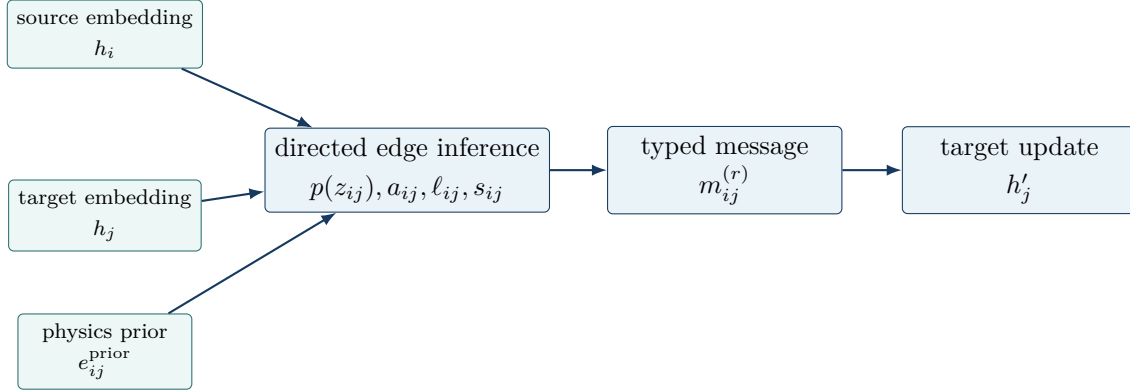


Figure 3: Directed physical-edge inference and message passing. The model first estimates the relation from source to target, then uses that relation to form a typed message and update the target variable representation.

10 Major GNN architectures and their relevance

10.1 Graph convolutional networks

A common GCN layer is

$$\mathbf{H}^{(\ell+1)} = \sigma \left(\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2} \mathbf{H}^{(\ell)} \mathbf{W}^{(\ell)} \right), \quad (10.1)$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$. GCNs are useful baselines, but the standard form assumes a fixed adjacency and often treats directions or relation types too simply for Module 11.

10.2 GraphSAGE

GraphSAGE computes

$$\bar{\mathbf{h}}_j^{(\ell)} = \text{AGG} \{ \mathbf{h}_i^{(\ell)} : i \in \mathcal{N}(j) \}, \quad (10.2)$$

$$\mathbf{h}_j^{(\ell+1)} = \sigma \left(\mathbf{W}^{(\ell)} [\mathbf{h}_j^{(\ell)} \parallel \bar{\mathbf{h}}_j^{(\ell)}] \right). \quad (10.3)$$

Its inductive formulation can generalize to graphs containing additional defect species or coefficient nodes not seen in every training sample.

10.3 Graph attention networks

A GAT uses attention coefficients

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(j)} \exp(e_{kj})}. \quad (10.4)$$

Attention is useful for state-dependent edge strength, but attention weights are not automatically causal or physically identifiable. They must be tested by interventions and predictive ablations.

10.4 Typed message-passing networks

A relational or typed MPNN uses a separate transformation for each physical relation:

$$\bar{\mathbf{m}}_j = \sum_r \sum_{i \in \mathcal{N}_r(j)} \phi_m^{(r)}(\mathbf{h}_i, \mathbf{h}_j, \mathbf{e}_{ij}). \quad (10.5)$$

This is the recommended architecture family because it matches the heterogeneous edge taxonomy.

10.5 Neural relational inference

NRI uses an encoder to infer latent edge types from observed trajectories and a decoder to predict dynamics conditioned on the inferred graph. This matches the core Module 11 problem. Strong physical priors can restrict the candidate edge set and assign interpretable edge categories instead of anonymous relation labels.

11 Temporal graphs and dynamical dependence

11.1 Instantaneous and lagged edges

The graph should separate two classes:

$$X_i(t) \rightarrow X_j(t) \quad \text{zero-lag algebraic or constitutive edge,} \quad (11.1)$$

$$X_i(t) \rightarrow X_j(t + \ell \Delta t) \quad \text{lagged dynamical edge.} \quad (11.2)$$

A single static adjacency matrix loses this distinction. One solution is a family of lag matrices

$$\mathbf{A}^{(0)}, \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(L_{\max})}. \quad (11.3)$$

11.2 Recurrent graph model

Let $\mathbf{z}_i^n$ be the compressed state of variable i at time t_n . A recurrent graph decoder can use

$$\mathbf{z}_j^{n+1} = F_j \left(\mathbf{z}_j^n, \sum_i \sum_{\ell=0}^{L_{\max}} A_{ij}^{(\ell)} M_{ij}^{(\ell)}(\mathbf{z}_i^{n-\ell}, \mathbf{z}_j^n), \mathbf{g}^n \right). \quad (11.4)$$

The model must use causal masking so that future data cannot influence the inferred past graph.

11.3 State-dependent graph strength

Topology may remain fixed while edge strength varies:

$$a_{ij}(t) = \text{sigmoid}(g_{ij}(\mathbf{h}_i(t), \mathbf{h}_j(t), \mathbf{g}(t))). \quad (11.5)$$

For example, Joule-heating feedback may be negligible at low bias and strong at high current. Defect-charge effects may be negligible before irradiation and dominant after damage accumulation. Learning state-dependent weights is therefore more realistic than learning one constant matrix.

12 Learning graph structure

12.1 Fixed candidate graph, weighted graph, and latent graph

Three levels are possible:

1. **fixed topology**: all trusted edges are fixed and only strengths are learned;
2. **candidate topology**: a physics-defined superset is given and the model gates edges on or off;
3. **fully latent topology**: every directed pair is allowed.

The second option is recommended. The first cannot discover omitted candidate mechanisms. The third permits too many spurious edges and requires far more data.

12.2 Continuous edge gates

A relaxed gate can be written

$$a_{ij} = \text{sigmoid}(\alpha_{ij}). \quad (12.1)$$

Sparsity is encouraged with

$$\mathcal{L}_{\text{sparse}} = \sum_{i \neq j} |a_{ij}|. \quad (12.2)$$

A gate close to zero suppresses messages but remains differentiable.

12.3 Discrete edge types

For categorical edge types, Gumbel-softmax provides a differentiable approximation to a discrete sample:

$$\tilde{z}_{ij,r} = \frac{\exp((\log \pi_{ij,r} + g_r)/\tau_g)}{\sum_s \exp((\log \pi_{ij,s} + g_s)/\tau_g)}. \quad (12.3)$$

The temperature τ_g is gradually reduced during training. A hard edge assignment should be evaluated at test time.

12.4 Causal discovery is not automatic

Predictive association does not prove causation. Two variables can correlate because of a common driver. For example, C and T can both rise after a radiation pulse even if neither directly drives the other in a simplified generator. Directionality must be supported by temporal precedence, conditional prediction, interventions, and physical equations.

12.5 Interventions

An intervention changes one mechanism while holding the rest of the generator fixed. If defect-dependent mobility is disabled, the response difference isolates the path

$$C \rightarrow \mu_{n,p} \rightarrow \mathbf{J}. \quad (12.4)$$

Intervention pairs provide much stronger supervision than passive trajectories.

13 Proposed Module 11 architecture

13.1 Node definition

The first model should use approximately 15–30 variable nodes. A recommended reduced node set is

$$\{C, T, D_C, \rho, \phi, \mathbf{E}, n, p, \mu_n, \mu_p, D_n, D_p, \tau_n, \tau_p, \mathbf{J}, G, Q_{\text{rad}}, Q_{\text{J}}\}. \quad (13.1)$$

A later model can separate defect species, electron and hole currents, reaction rates, and heat-source components.

13.2 Field-to-node feature extraction

Although many nodes represent spatial fields, the graph is not spatial. Each field is summarized by a feature vector such as

$$\mathbf{x}_i(t) = \left[\overline{X_i}, \sigma(X_i), \min X_i, \max X_i, \|X_i\|_2, \|\nabla X_i\|_2, q_{0.1}, q_{0.5}, q_{0.9}, \dot{\overline{X_i}}, a_{i,1}, \dots, a_{i,K} \right], \quad (13.2)$$

where $a_{i,k}$ are optional POD coefficients. These descriptors preserve field information while keeping graph nodes tied to physical variables.

13.3 Node-type encoder

Different variable types require different encoders:

$$\mathbf{h}_i^0 = \text{Enc}_{\tau(i)}(\mathbf{x}_i), \quad (13.3)$$

where $\tau(i)$ denotes field, coefficient, source, or rate node. Type-specific encoders avoid asking one normalization and one MLP to handle incompatible semantics.

13.4 Edge prior mask

Define a candidate mask $M_{ij} \in \{0, 1\}$. The edge logit is

$$\alpha_{ij} \leftarrow \begin{cases} \alpha_{ij}, & M_{ij} = 1, \\ -\infty, & M_{ij} = 0. \end{cases} \quad (13.4)$$

Trusted impossible edges are excluded. Plausible but uncertain edges remain learnable. Candidate masks should be versioned because they encode scientific assumptions.

13.5 Encoder-decoder structure

The recommended model is:

1. encode node trajectories over a context window;
2. infer directed edge probabilities, types, signs, and lags;
3. pass typed messages through the inferred graph;
4. decode future node states;
5. compute physics and intervention losses;
6. derive SCCs and the condensation sequence from the thresholded graph.

13.6 Graph output and thresholding

At inference time, report continuous probabilities first. A binary graph is obtained using a validation-selected threshold τ_A :

$$\hat{A}_{ij} = \mathbb{I}[p(i \rightarrow j) > \tau_A]. \quad (13.5)$$

The threshold should maximize a balanced graph-recovery metric on validation data, not be chosen after inspecting the test set.

14 Synthetic data: what it can and cannot establish

14.1 What synthetic data can prove

Synthetic data are sufficient to test whether the method can recover a known graph, distinguish direct from indirect dependence, identify edge direction and type, estimate physical lags, track regime-dependent strengths, generalize to held-out parameter combinations, and predict the response to mechanism ablations.

14.2 What synthetic data cannot prove

The learned graph is only as complete as the generator family. If the simulator omits an interface trap mechanism, nonlocal phonon effect, field-assisted reaction, or material transition, the network cannot discover it from data that never contain it. Therefore,

$$\text{recovered graph} \approx \text{best graph inside the synthetic model family}, \quad (14.1)$$

not necessarily the complete graph of a real device.

14.3 Avoiding circularity

The graph labels must come from equations and explicit mechanism switches, not from the hand-drawn Module 6 diagram alone. The simulator should expose every candidate edge through a coupling coefficient or mechanism flag. Synthetic cases should include multiple graph topologies, not one fixed topology repeated under different parameters. Otherwise, the network can memorize the graph without learning how evidence supports it.

15 Synthetic sample definition

One synthetic sample should contain

$$\mathcal{D}^{(s)} = \left\{ \mathcal{P}^{(s)}, \mathcal{B}^{(s)}, \mathcal{I}^{(s)}, \mathcal{S}^{(s)}, \mathcal{K}^{(s)}, \mathcal{X}^{(s)}(t), \mathcal{Y}^{(s)}(t), \mathbf{A}_{\text{true}}^{(s)}, \mathbf{Z}_{\text{true}}^{(s)}, \mathbf{L}_{\text{true}}^{(s)} \right\}. \quad (15.1)$$

Here:

- $\mathcal{P}$ contains physical parameters;
- $\mathcal{B}$ contains boundary and forcing conditions;
- $\mathcal{I}$ contains initial conditions;
- $\mathcal{S}$ contains source realizations;
- $\mathcal{K}$ contains active mechanism switches and edge multipliers;
- $\mathcal{X}(t)$ contains raw field trajectories;
- $\mathcal{Y}(t)$ contains normalized variable-node features;
- $\mathbf{A}_{\text{true}}$ is the known directed graph;
- $\mathbf{Z}_{\text{true}}$ stores edge types and signs;
- $\mathbf{L}_{\text{true}}$ stores physical lags or lag classes.

16 Synthetic generator architecture

16.1 Layer 1: graph-family sampler

Begin with a trusted core graph and a set of switchable candidate edges. For each sample, draw mechanism switches

$$k_e \in \{0, 1\} \quad (16.1)$$

or continuous edge multipliers

$$\gamma_e \in [0, 1]. \quad (16.2)$$

Examples include:

$$\gamma_{C \rightarrow \rho} : \text{effective defect charge}, \quad (16.3)$$

$$\gamma_{C \rightarrow \mu} : \text{defect mobility degradation}, \quad (16.4)$$

$$\gamma_{C \rightarrow \tau} : \text{defect-assisted recombination}, \quad (16.5)$$

$$\gamma_{E \rightarrow C} : \text{charged-defect drift}, \quad (16.6)$$

$$\gamma_{J, E \rightarrow T} : \text{Joule heating}, \quad (16.7)$$

$$\gamma_{R \rightarrow T} : \text{recombination heating}, \quad (16.8)$$

$$\gamma_{T \rightarrow R_j} : \text{thermally activated defect reactions}. \quad (16.9)$$

The ground-truth adjacency is generated directly from active switches.

16.2 Layer 2: parameter sampler

Parameters should be sampled jointly rather than one at a time. Use a mixture of:

- uniform sampling for bounded approximately linear uncertainty;
- log-uniform sampling for positive quantities spanning orders of magnitude;
- beta distributions for values concentrated near a nominal range;
- categorical sampling for material, boundary, or mechanism families;
- Latin hypercube or Sobol designs for efficient multidimensional coverage;
- targeted stress samples near known transitions or stiffness boundaries.

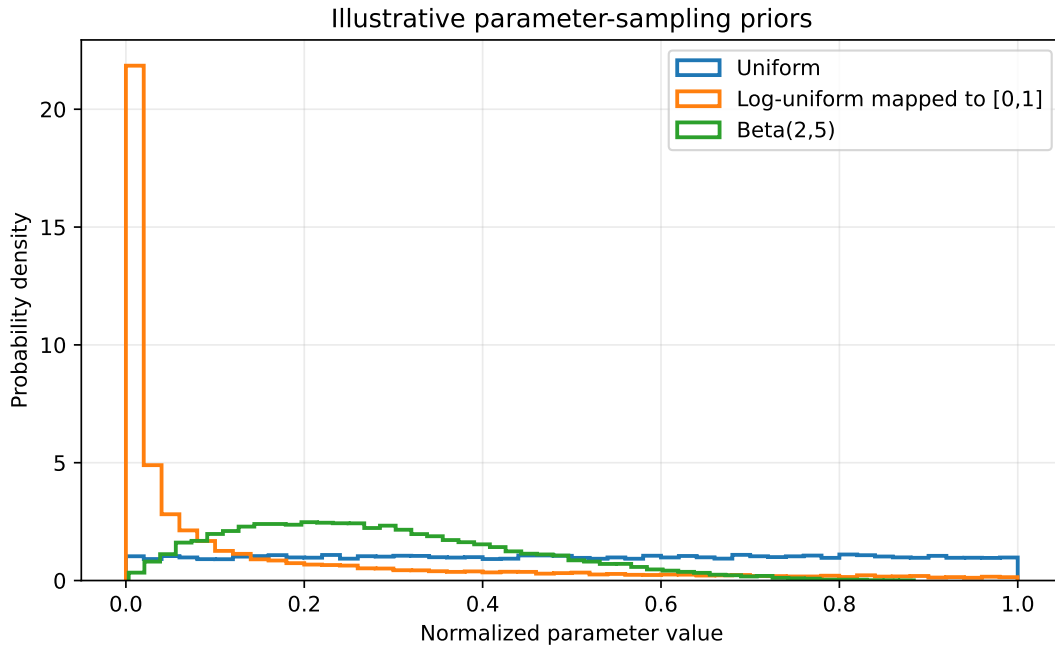


Figure 4: Illustrative sampling priors. Uniform, log-uniform, and beta distributions serve different purposes. The plotted curves are conceptual and not calibrated project distributions.

16.3 Layer 3: source and initial-condition sampler

Radiation-induced defect and heat sources should vary in amplitude, location, width, shape, and temporal profile. A Gaussian source is

$$S_C(x, y, t) = A_C \exp \left[-\frac{(x - x_0)^2}{2\sigma_x^2} - \frac{(y - y_0)^2}{2\sigma_y^2} \right] f_C(t). \quad (16.10)$$

Useful temporal functions include:

$$f_{\text{imp}}(t) \approx \exp \left[-(t - t_0)^2 / (2\sigma_t^2) \right], \quad (16.11)$$

$$f_{\text{step}}(t) = H(t - t_0), \quad (16.12)$$

$$f_{\text{decay}}(t) = H(t - t_0) e^{-(t-t_0)/\tau_s}, \quad (16.13)$$

$$f_{\text{pulse}}(t) = H(t - t_0) - H(t - t_1). \quad (16.14)$$

A smooth random field can be created through a truncated Karhunen–Loève expansion:

$$X(x, y) = \bar{X}(x, y) + \sum_{k=1}^K \sqrt{\lambda_k} \xi_k \psi_k(x, y), \quad \xi_k \sim \mathcal{N}(0, 1). \quad (16.15)$$

This varies field morphology without changing the graph topology.

16.4 Layer 4: mechanistic trajectory generator

Run the coupled physics equations with the sampled graph, parameters, sources, and initial conditions. The generator must save raw states at every requested physical time. The numerical method is not a target; only the resulting trajectories and residuals are used.

16.5 Layer 5: intervention generator

For a selected baseline sample, create matched counterfactuals by changing one mechanism:

$$(\mathcal{P}, \mathcal{B}, \mathcal{I}, \mathcal{S})_{\text{baseline}} = (\mathcal{P}, \mathcal{B}, \mathcal{I}, \mathcal{S})_{\text{intervention}}, \quad (16.16)$$

while k_e or γ_e changes. The matched design isolates the edge effect.

16.6 Layer 6: observation and noise model

Create several observation levels:

1. exact simulator fields;

2. field summaries with no added noise;
3. multiplicative and additive noise;
4. missing variables at selected times;
5. coarser temporal sampling;
6. model-discrepancy cases generated by a richer equation set.

For positive quantities, multiplicative noise is natural:

$$X_{\text{obs}} = X \exp(\sigma_\eta \eta), \quad \eta \sim \mathcal{N}(0, 1). \quad (16.17)$$

For signed quantities, use additive noise scaled by a robust range.

17 Recommended synthetic parameter ranges

The following ranges are initial stress-test ranges, not calibrated device specifications. They should be narrowed when material-specific data become available.

Quantity	Illustrative range	Distribution		Purpose
Ambient temperature T_0	250–400 K	uniform or beta		Baseline thermal state and coefficient variation.
Transient temperature rise	10^{-3} – 10^2 K	log-uniform		Weak to strong heating.
Applied bias magnitude	10^{-3} – 10^2 V	signed	log-uniform	Carrier-electrostatic feedback strength.
Peak effective defect density C_{max}	10^{14} – 10^{24} m $^{-3}$	log-uniform		Pristine to heavily damaged regimes.
Defect charge magnitude $ z_{\text{def}} $	0–2	categorical	or continuous	Activates or weakens $C \rightarrow \rho$.
Defect source amplitude A_C	problem-scaled over 6–10 decades	log-uniform		Radiation-damage intensity.
Defect diffusivity prefactor $D_{0,C}$	10^{-20} – 10^{-4} m 2 /s	log-uniform		Frozen to mobile defects.
Defect activation energy E_a	0.05–2.0 eV	uniform/beta		Thermal sensitivity of defect motion and reaction.
Annealing time scale	10^{-9} – 10^8 s	log-uniform		Fast recovery to long-term damage.
Electron mobility $\mu_{n,0}$	10^{-4} – 1 m 2 /(V s)	log-uniform		Carrier drift strength.
Hole mobility $\mu_{p,0}$	10^{-5} – 0.5 m 2 /(V s)	log-uniform		Hole drift strength.
Mobility temperature exponent	0–3	uniform		Controls $T \rightarrow \mu$.

Quantity		Illustrative range	Distribution	Purpose
Defect mobility-degradation factor		0–10 ³ normalized	log-uniform plus zero mass	Controls $C \rightarrow \mu$.
Carrier lifetime τ_n, τ_p		10 ⁻¹² –10 ⁻² s	log-uniform	Recombination time scales.
Carrier generation G		problem-scaled over 8 decades	log-uniform plus zero mass	Radiation or external carrier source.
Thermal conductivity k		0.1–500 W/(m K)	log-uniform	Heat-removal strength.
Volumetric heat capacity $\rho_m c_p$		10 ⁴ –10 ⁷ J/(m ³ K)	log-uniform	Thermal response time.
Radiation heat source Q_{rad}		problem-scaled over 10 decades	log-uniform plus zero mass	Direct thermal forcing.
Joule-heating multiplier γ_J	multiplier	0–1.5	mixture with mass at zero and one	Edge activation and model mismatch.
Recombination-heating multiplier		0–1.5	mixture	Tests additional heat pathway.
Edge lag class ℓ_{ij}		0–8 stored time steps	categorical	Lag recovery.
Observation noise		0–10% RMS	mixture	Robustness and uncertainty calibration.

17.1 Correlated sampling

Some parameters should not be independent. Examples include the Einstein relation

$$D_n = \frac{k_B T}{q} \mu_n, \quad D_p = \frac{k_B T}{q} \mu_p, \quad (17.1)$$

and Arrhenius diffusivity

$$D_C(T) = D_{0,C} \exp\left(-\frac{E_a}{k_B T}\right). \quad (17.2)$$

Sample the independent base parameters, then compute dependent coefficients. Violating such relations without labeling the case as a discrepancy study teaches the GNN physically impossible combinations.

17.2 Dimensionless regime sampling

Raw parameter ranges do not guarantee coverage of physically distinct regimes. Stratify samples using dimensionless groups such as

$$\text{Pe}_n = \frac{\mu_n |\mathbf{E}| L}{D_n}, \quad (17.3)$$

$$\text{Da}_C = \frac{L^2}{D_C \tau_{\text{react}}}, \quad (17.4)$$

$$\Gamma_J = \frac{Q_J L^2}{k \Delta T_{\text{ref}}}, \quad (17.5)$$

$$\Gamma_\rho = \frac{q|z|CL^2}{\epsilon \Delta \phi_{\text{ref}}}. \quad (17.6)$$

Balanced coverage of low, intermediate, and high values helps the network observe when edges become weak or dominant.

18 Detailed intervention library

18.1 Hard edge removal

Set the mechanism coefficient exactly to zero. Examples are

$$z_{\text{def}} = 0 \quad \Rightarrow C \not\rightarrow \rho, \quad (18.1)$$

$$\alpha_{\mu C} = 0 \quad \Rightarrow C \not\rightarrow \mu_{n,p}, \quad (18.2)$$

$$\alpha_{\tau C} = 0 \quad \Rightarrow C \not\rightarrow \tau_{n,p}, \quad (18.3)$$

$$\gamma_J = 0 \quad \Rightarrow (\mathbf{J}, \mathbf{E}) \not\rightarrow T, \quad (18.4)$$

$$\mu_C = 0 \quad \Rightarrow \mathbf{E} \not\rightarrow C \text{ through charged-defect drift.} \quad (18.5)$$

18.2 Soft intervention

Scale an edge by $\gamma_e \in \{0.1, 0.25, 0.5, 0.75, 1.0, 1.5\}$. A correct model should infer a monotonic increase in effective edge strength when the underlying mechanism is monotonic.

18.3 Node intervention

Clamp a variable to a prescribed trajectory:

$$\text{do}(T(t) = T_{\text{clamp}}(t)). \quad (18.6)$$

This breaks incoming edges to T while preserving outgoing temperature effects. Node interventions help distinguish incoming from outgoing causation.

18.4 Source intervention

Apply isolated forcing to one source at a time: defect-only, carrier-only, heat-only, or bias-only excitation. Independent excitation improves graph identifiability because variables no longer move

together solely due to one common radiation pulse.

18.5 Factorial intervention design

Single-edge ablations do not reveal interactions between mechanisms. A fractional factorial design over key switches can test whether two pathways mask or amplify each other. At minimum, include combinations of defect charge, defect-mobility degradation, defect-assisted recombination, and Joule heating.

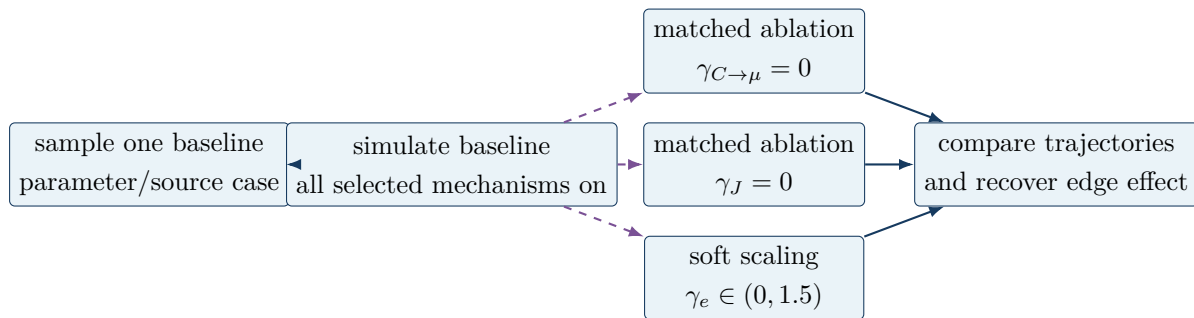


Figure 5: Matched intervention generation. The baseline and counterfactuals share parameters, sources, and initial conditions; only one physical mechanism changes.

19 Synthetic dataset generation algorithm

Algorithm: generate one graph-labeled synthetic family

1. Draw a graph family by activating the trusted core edges and sampling candidate mechanism switches.
2. Draw independent base parameters using a stratified design.
3. Enforce constitutive and dimensional constraints to obtain dependent coefficients.
4. Draw radiation sources, bias, boundary conditions, and initial fields.
5. Integrate the physical equations and save all variables at requested times.
6. Compute raw and normalized node features.
7. Store the exact directed adjacency, edge types, signs, lags, and mechanism multipliers.
8. Generate matched hard and soft interventions for selected edges.
9. Compute equation residuals and conservation diagnostics.
10. Reject only numerically invalid cases using predeclared criteria; do not remove physically difficult cases simply because they are inconvenient.

19.1 Pseudocode

```

for graph_family = 1:N_graph_families
    mechanism = sample_mechanism_switches(candidate_edges);
    A_true, edge_type, edge_lag = build_ground_truth_graph(mechanism);

    for case_id = 1:N_cases_per_family
        base = sample_independent_parameters(design);
        params = enforce_physical_constraints(base, mechanism);
        source = sample_radiation_and_bias_sources(params);
        initial = sample_initial_conditions(params, source);

        trajectory = simulate_physical_case(params, source, initial,
            mechanism);
        features = extract_variable_node_features(trajectory);
        diagnostics = compute_physics_residuals(trajectory, params);

        save_case(features, trajectory, diagnostics,
            A_true, edge_type, edge_lag, mechanism);
    end
end

```

```

        for intervention = selected_interventions(mechanism)
            counterfactual = simulate_matched_intervention(
                params, source, initial, mechanism, intervention);
            save_intervention_pair(trajjectory, counterfactual,
                intervention);
        end
    end
end

```

19.2 Dataset size ladder

A practical progression is:

1. 100–300 toy ODE cases for debugging direction and lag;
2. 500–2,000 lumped multiphysics cases for architecture selection;
3. 2,000–10,000 reduced field-simulation cases for proof of concept;
4. a smaller high-fidelity discrepancy set for robustness testing.

Each baseline case can produce several intervention pairs, substantially increasing causal supervision.

19.3 Active sampling

After an initial model is trained, add cases where edge entropy is high:

$$H_{ij} = - \sum_r p(z_{ij} = r) \log p(z_{ij} = r). \quad (19.1)$$

Also target physical regimes in which two candidate graphs make similar passive predictions but different intervention predictions.

20 Training targets and losses

20.1 Multi-task objective

A recommended objective is

$$\mathcal{L}_{\text{total}} = \lambda_x \mathcal{L}_{\text{state}} + \lambda_e \mathcal{L}_{\text{edge}} + \lambda_t \mathcal{L}_{\text{type}} + \lambda_\ell \mathcal{L}_{\text{lag}} + \lambda_p \mathcal{L}_{\text{physics}} + \lambda_c \mathcal{L}_{\text{cons}} + \lambda_i \mathcal{L}_{\text{int}} + \lambda_s \mathcal{L}_{\text{sparse}} + \lambda_r \mathcal{L}_{\text{prior}}. \quad (20.1)$$

20.2 State-prediction loss

For normalized node states,

$$\mathcal{L}_{\text{state}} = \frac{1}{N_t N_v} \sum_{n,j} \left\| \hat{\mathbf{z}}_j^{n+1} - \mathbf{z}_j^{n+1} \right\|_2^2. \quad (20.2)$$

Multi-step rollout loss should also be included:

$$\mathcal{L}_{\text{state}}^{(K)} = \sum_{k=1}^K w_k \left\| \hat{\mathbf{z}}^{n+k} - \mathbf{z}^{n+k} \right\|_2^2. \quad (20.3)$$

A graph that predicts only one step but drifts rapidly is not sufficient.

20.3 Edge-existence loss

For binary edge labels,

$$\mathcal{L}_{\text{edge}} = - \sum_{i \neq j} [y_{ij} \log p_{ij} + (1 - y_{ij}) \log(1 - p_{ij})]. \quad (20.4)$$

Because absent edges outnumber present edges, use class weighting or focal loss.

20.4 Edge-type and direction loss

For relation type r ,

$$\mathcal{L}_{\text{type}} = - \sum_{i \neq j} \sum_r y_{ijr} \log p_{ijr}. \quad (20.5)$$

Direction is assessed through separate ordered pairs. A prediction of $j \rightarrow i$ does not partially satisfy a true edge $i \rightarrow j$.

20.5 Lag loss

For categorical lag classes,

$$\mathcal{L}_{\text{lag}} = - \sum_{i,j,\ell} y_{ij\ell} \log p_{ij\ell}. \quad (20.6)$$

For continuous delay estimates, use a robust regression loss such as Huber loss.

20.6 Physics residual losses

Predicted trajectories should satisfy reduced governing relations. Examples include

$$\mathcal{R}_\rho = \rho - q \left[p - n + N_D^+ - N_A^- + \sum_j z_j C_j \right], \quad (20.7)$$

$$\mathcal{R}_\phi = -\nabla \cdot (\epsilon \nabla \phi) - \rho, \quad (20.8)$$

$$\mathcal{R}_E = \mathbf{E} + \nabla \phi, \quad (20.9)$$

$$\mathcal{R}_{J_n} = \mathbf{J}_n - q \mu_n n \mathbf{E} - q D_n \nabla n, \quad (20.10)$$

$$\mathcal{R}_T = \rho_m c_p \partial_t T - \nabla \cdot (k \nabla T) - Q_{\text{tot}}. \quad (20.11)$$

Then

$$\mathcal{L}_{\text{physics}} = \sum_q w_q \frac{\|\mathcal{R}_q\|_2^2}{s_q^2 + \varepsilon}. \quad (20.12)$$

The scales s_q prevent one equation from dominating solely because of units.

20.7 Conservation losses

Integrated carrier consistency can be enforced through

$$\frac{d}{dt} \int_\Omega n \, d\Omega - \int_\Omega (G - R) \, d\Omega - \frac{1}{q} \int_{\partial\Omega} \mathbf{J}_n \cdot \hat{\mathbf{n}} \, d\Gamma = 0, \quad (20.13)$$

with analogous hole, defect, and energy balances.

20.8 Intervention consistency

For a matched baseline and intervention,

$$\Delta \mathbf{z}^{\text{true}} = \mathbf{z}^{\text{int}} - \mathbf{z}^{\text{base}}, \quad (20.14)$$

require

$$\mathcal{L}_{\text{int}} = \left\| \Delta \hat{\mathbf{z}} - \Delta \mathbf{z}^{\text{true}} \right\|_2^2. \quad (20.15)$$

This loss directly trains the model to predict mechanism-specific effects.

20.9 Prior consistency and forbidden edges

Hard-impossible edges are masked. Trusted edges can receive a soft prior:

$$\mathcal{L}_{\text{prior}} = \sum_{(i,j) \in \mathcal{E}_{\text{trusted}}} (1 - p_{ij})^2 + \sum_{(i,j) \in \mathcal{E}_{\text{unlikely}}} \eta_{ij} p_{ij}^2. \quad (20.16)$$

A soft prior allows the model to expose simulator inconsistencies rather than forcing agreement at all costs.

20.10 Loss-weight selection

Losses should be normalized, monitored separately, and tuned using validation performance. A curriculum can first train state prediction with a fixed graph, then activate edge learning, then add intervention and sparsity losses. Dynamically balancing gradients is preferable to arbitrary weights spanning many orders of magnitude.

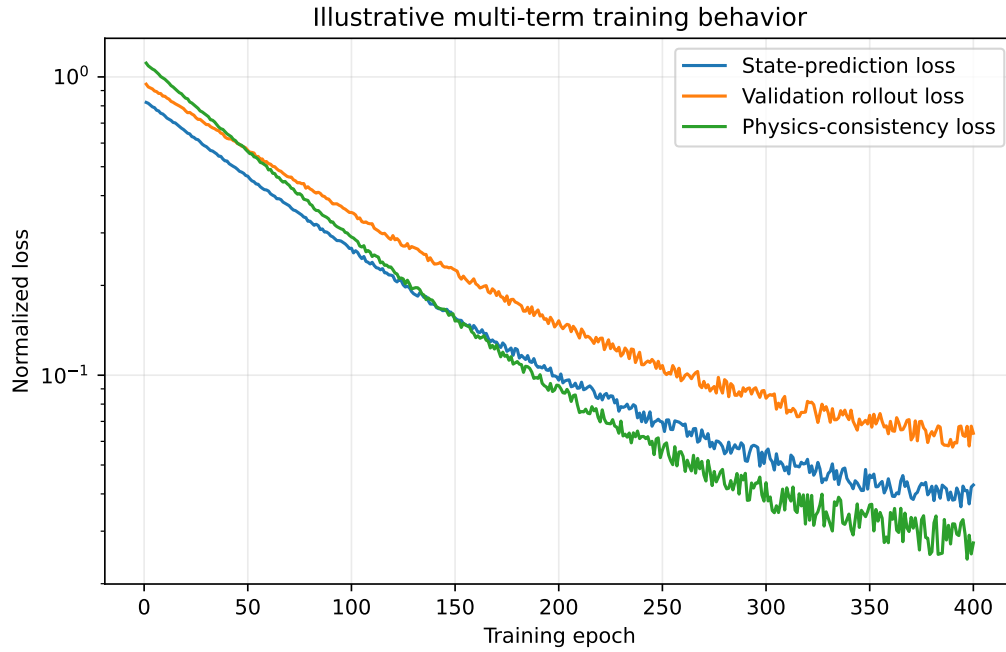


Figure 6: Illustrative multi-term training behavior. These curves are conceptual and do not represent a trained Module 11 model. In the revised scope, the validation rollout loss measures physical-trajectory prediction rather than numerical schedule quality.

21 Training curriculum

21.1 Phase 0: analytic two- and three-node systems

Verify edge orientation, lag handling, and intervention logic on systems with exact solutions. Examples include a chain $x \rightarrow y \rightarrow z$, a common-cause structure $x \leftarrow u \rightarrow y$, and a two-node delayed feedback loop.

21.2 Phase 1: fixed graph, state prediction

Provide the true adjacency and train only the dynamics decoder. Failure here indicates inadequate node features or model capacity, not graph-inference failure.

21.3 Phase 2: edge strength

Keep topology fixed but learn continuous mechanism multipliers. Verify that predicted strength tracks known γ_e .

21.4 Phase 3: edge presence and type

Allow switchable candidate edges. Train on multiple graph families and evaluate exact topology recovery.

21.5 Phase 4: lag and feedback groups

Add time-lag classification and SCC analysis. Verify the condensation sequence against the known graph.

21.6 Phase 5: noisy and discrepant data

Add noise, missing variables, coarser time sampling, and richer hidden physics. The objective is calibrated uncertainty, not forced confident recovery.

22 Train, validation, test, and OOD splits

22.1 Split by complete physical family

All time steps, interventions, and source variants associated with one base case should remain in one split. Otherwise, nearly identical counterfactuals leak into training and test data.

22.2 Topology split

Hold out selected combinations of active candidate edges. This tests whether the model composes known mechanisms rather than memorizing complete graphs.

22.3 Parameter extrapolation

Reserve high-damage, high-bias, or extreme-temperature ranges for OOD testing. Report exactly which dimensions are outside the training support.

22.4 Mechanism OOD

Generate a test set containing an unmodeled pathway, such as recombination heating, while excluding that edge from the candidate graph. A reliable model should show elevated residuals or uncertainty rather than silently misattribute the effect.

22.5 Source-shape transfer

Train on Gaussian and rectangular source profiles and test on clustered or multi-spot sources. The graph should remain stable even when field morphology changes.

23 Evaluation metrics

23.1 Graph topology

Report directed edge precision, recall, and F1 score:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (23.1)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (23.2)$$

$$F_1 = \frac{2 \text{Precision Recall}}{\text{Precision} + \text{Recall}}. \quad (23.3)$$

Also report structural Hamming distance, AUROC, and area under the precision-recall curve.

23.2 Direction and type

Conditioned on a detected adjacency, report direction accuracy and edge-type accuracy. A reversed edge should count as a structural error even if the unordered pair is correct.

23.3 Lag

Report lag-class accuracy, mean absolute lag error, and uncertainty calibration. Zero-lag algebraic edges should be assessed separately from delayed dynamic edges.

23.4 Edge strength

For known continuous mechanism multipliers, report correlation, mean absolute error, and monotonicity violations between γ_e and predicted a_{ij} .

23.5 Trajectory and intervention prediction

Report normalized one-step error, rollout error, intervention-effect error, and residual norms. A graph with good F1 but poor counterfactual prediction is not physically useful.

23.6 SCC and sequence recovery

Compare the inferred SCC partition with the true partition using adjusted Rand index or pairwise agreement. Compare the condensation-graph partial order rather than forcing an arbitrary order within feedback groups.

23.7 Calibration

For predicted edge probability p , empirical correctness should be approximately p . Reliability diagrams, expected calibration error, and Brier scores should be reported.

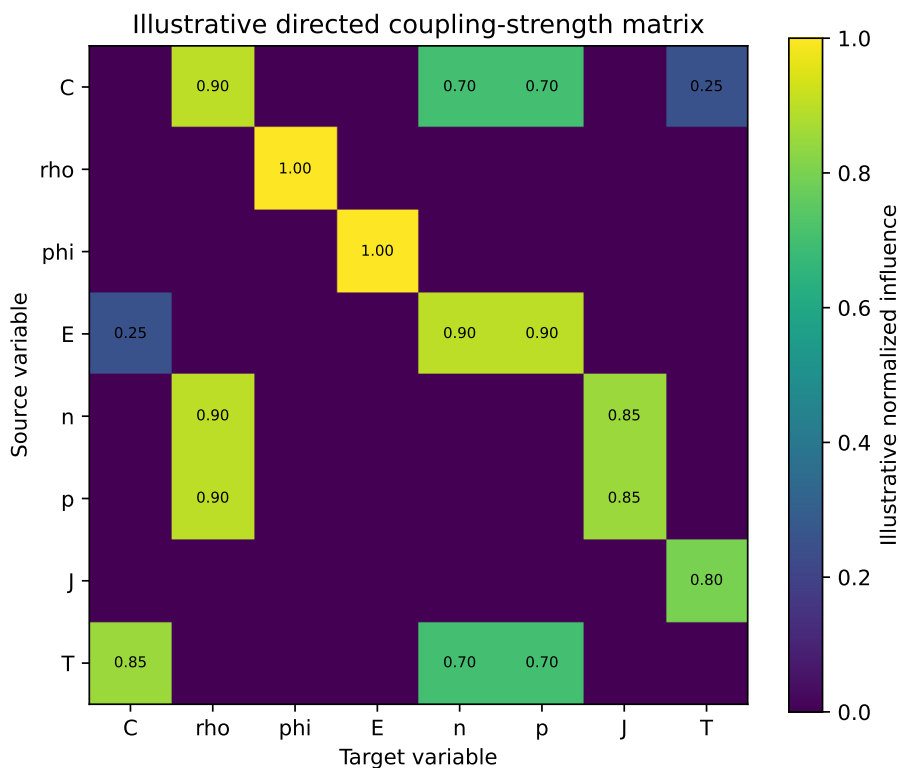


Figure 7: Illustrative directed coupling-strength matrix. Rows are source variables and columns are targets. The values are conceptual and should not be interpreted as project results.

24 Baselines the GNN must beat

A GNN should be compared against simpler structure-learning methods:

- pairwise correlation and lagged cross-correlation;
- partial correlation conditioned on candidate mediators;
- linear vector autoregression and Granger-style tests;
- sparse regression on candidate nonlinear library terms;
- an MLP over concatenated variable features with pairwise edge heads;
- fixed hand-coded graph with learned edge strengths;
- neural relational inference without physics losses;
- ablations without intervention data or without relation types.

The strongest physics baseline is direct extraction from the known synthetic equations. The learned model is justified when it recovers that graph from observations and generalizes across graph families and regimes.

25 Proof-of-concept experiment ladder

25.1 Experiment 0: three-node chain

Use

$$\dot{x} = -ax + u(t), \quad (25.1)$$

$$\dot{y} = -by + cx, \quad (25.2)$$

$$\dot{z} = -dz + ey. \quad (25.3)$$

The graph is $x \rightarrow y \rightarrow z$. Verify that the model does not create a direct $x \rightarrow z$ edge when y is observed.

25.2 Experiment 1: common cause

Use

$$\dot{x} = -ax + cu, \quad (25.4)$$

$$\dot{y} = -by + du. \quad (25.5)$$

Passive trajectories correlate x and y , but the true graph is $u \rightarrow x$ and $u \rightarrow y$. This tests resistance to spurious association.

25.3 Experiment 2: delayed feedback

Use

$$\dot{x}(t) = -ax(t) + by(t - \tau_y), \quad (25.6)$$

$$\dot{y}(t) = -cy(t) + dx(t - \tau_x). \quad (25.7)$$

Verify direction, lag, SCC recovery, and time-unrolled representation.

25.4 Experiment 3: lumped Module 2–5 model

Replace spatial derivatives by low-order effective terms and retain variables $C, T, \rho, \phi, E, n, p, J$. Randomly activate graph mechanisms and generate thousands of inexpensive trajectories. This is the main architecture-selection environment.

25.5 Experiment 4: reduced field simulation

Generate field trajectories from the existing coupled physics modules, compress each physical field into variable-node descriptors, and recover the known graph. The graph topology remains variable based; no spatial adjacency is learned.

25.6 Experiment 5: hidden-mechanism discrepancy

Generate test trajectories with one pathway omitted from the candidate graph. Evaluate whether residuals and uncertainty identify inadequacy.

26 Uncertainty and safe interpretation

26.1 Sources of uncertainty

Uncertainty arises from finite data, overlapping effects, insufficient excitation, observation noise, hidden variables, model discrepancy, and neural optimization variability.

26.2 Practical methods

Use deep ensembles, Monte Carlo dropout, bootstrap resampling by physical case, and Bayesian or Dirichlet edge heads. Report edge probabilities and confidence intervals rather than only a thresholded graph.

26.3 Abstention

When the model is uncertain, it should return an unresolved edge or a set of plausible graphs. Scientific structure learning is harmed by mandatory hard decisions.

27 Interpretability

27.1 Edge probability is not enough

For every inferred edge, report supporting evidence:

- trajectory prediction improvement when the edge is included;
- degradation when the edge is ablated;
- intervention response agreement;
- physical residual reduction;
- edge probability across ensemble members;
- state regimes where the edge becomes strong.

27.2 Counterfactual explanation

For an inferred edge $i \rightarrow j$, compare the predicted target trajectory with and without that edge while holding other inferred relations fixed:

$$\Delta \hat{X}_j^{(i \rightarrow j)}(t) = \hat{X}_j^{\text{full}}(t) - \hat{X}_j^{\text{edge removed}}(t). \quad (27.1)$$

This provides a mechanistic effect estimate.

27.3 Regime maps

Plot edge strength against physically meaningful coordinates such as defect density, temperature, electric field, bias, and dimensionless groups. The result is a map of when a coupling matters rather than a single global number.

28 Computational considerations

The variable graph is small, so GNN inference cost is negligible compared with synthetic trajectory generation. The dominant expense is producing diverse coupled simulations and interventions. Data generation should therefore be parallelized by independent physical case. Store compressed node features for training but retain a reproducible subset of raw trajectories for diagnostics.

The pairwise edge-inference cost scales as

$$O(N_v^2 d_h) \quad (28.1)$$

for a dense candidate graph, or

$$O(N_e d_h) \quad (28.2)$$

for a masked candidate set. With $N_v \lesssim 30$, either is modest.

29 MATLAB-oriented data representation

29.1 Recommended case structure

```
caseData.id
caseData.graphFamily
caseData.params
caseData.sources
caseData.initialConditions
caseData.mechanismSwitches
caseData.edgeMultipliers
caseData.time
caseData.rawFields
caseData.nodeFeatures
caseData.nodeTypes
caseData.candidateMask
caseData.trueAdjacency
caseData.trueEdgeTypes
caseData.trueEdgeSigns
caseData.trueEdgeLags
caseData.physicsResiduals
caseData.conservativeDiagnostics
caseData.interventionPairs
caseData.randomSeed
caseData.generatorVersion
```

29.2 Node-feature schema

Store one tensor with shape

$$N_{\text{case}} \times N_t \times N_v \times N_f. \quad (29.1)$$

Use masks for variables unavailable in a given graph family. Node type and units should be stored separately rather than inferred from column position.

29.3 Edge-label schema

Store ordered-pair labels with shape

$$N_{\text{case}} \times N_v \times N_v. \quad (29.2)$$

Additional tensors store relation type, sign, lag, and mechanism multiplier. Self-edges should be either excluded or explicitly defined as persistence edges.

29.4 MATLAB versus Python training

The project can generate and validate physics trajectories in MATLAB. The GNN can be implemented with MATLAB Deep Learning Toolbox using custom `dlnetwork` layers, or trained in Python after export to HDF5. The data schema should remain language neutral. A first proof of concept should prefer the environment that makes directed typed message passing and uncertainty evaluation easiest, while keeping the authoritative physics generator in MATLAB.

30 Proposed Module 11 file architecture

Planned file	Purpose
<code>matlab/main_module11_generate_dependency_dataset.m</code>	Generate graph families, parameters, trajectories, interventions, and labels.
<code>matlab/main_module11_train_dependency_gnn.m</code>	Train fixed-graph, edge-strength, and latent-edge models.
<code>matlab/main_module11_evaluate_dependency_graph.m</code>	Evaluate topology, direction, type, lag, SCCs, interventions, residuals, and calibration.
<code>matlab/src/module11_gnn/build_candidate_variable_graph.m</code>	Define nodes, candidate directed edges, trusted masks, and edge priors.
<code>matlab/src/module11_gnn/sample_graph_family.m</code>	Activate candidate physical mechanisms and create ground-truth labels.
<code>matlab/src/module11_gnn/sample_physical_parameters.m</code>	Draw constrained and stratified parameter sets.
<code>matlab/src/module11_gnn/sample_sources_initial_conditions.m</code>	Generate radiation sources, forcing, and initial fields.
<code>matlab/src/module11_gnn/simulate_physical_graph_case.m</code>	Run the mechanistic Modules 2–5/6 trajectory generator.
<code>matlab/src/module11_gnn/apply_physical_intervention.m</code>	Create matched hard, soft, node, and source interventions.

Planned file	Purpose
matlab/src/module11_gnn/extract_variable_node_features.m	Convert raw physical fields into variable-node descriptors or POD coefficients.
matlab/src/module11_gnn/compute_physics_graph_labels.m	Store adjacency, relation type, sign, lag, and SCC labels.
matlab/src/module11_gnn/compute_graph_physics_residuals.m	Evaluate equation and conservation residuals.
matlab/src/module11_gnn/train_graph_baselines.m	Correlation, Granger-style, sparse-regression, and MLP baselines.
matlab/src/module11_gnn/train_typed_dependency_mpnn.m	Train the directed typed GNN.
matlab/src/module11_gnn/calibrate_edge_uncertainty.m	Calibrate probabilities and ensemble uncertainty.
matlab/src/module11_gnn/derive_physical_coupling_sequence.m	Compute SCCs, condensation graph, and partial ordering.
matlab/tests/test_module11_direct_edge_recovery.m	Verify direct versus indirect edge recovery.
matlab/tests/test_module11_intervention_recovery.m	Verify mechanism ablation and soft-scaling response.
matlab/tests/test_module11_lag_and_scc_recovery.m	Verify lag classes, feedback groups, and condensation sequence.
matlab/tests/test_module11_hidden_mechanism_uncertainty.m	Verify uncertainty under candidate-graph misspecification.

31 Verification and acceptance criteria

A proof of concept should satisfy predeclared criteria such as:

1. recover at least 95% of edges in noise-free toy systems;
2. avoid direct shortcut edges when observed mediators explain the relation;
3. exceed non-graph baselines on held-out topology combinations;
4. recover edge direction and mechanism type separately;
5. recover zero-lag versus delayed classification;
6. predict matched intervention effects within a chosen normalized error tolerance;
7. reproduce true SCC partitions and condensation ordering;

8. maintain calibrated uncertainty under noise and missing variables;
9. show increased uncertainty or residuals for hidden-mechanism test cases;
10. reproduce results across at least five random training seeds.

Numerical thresholds for field-scale experiments should be set after baseline variability is measured, but before final test evaluation.

32 Common failure modes

32.1 Confusing association with direct dependence

A strong C - ϕ correlation does not imply a direct edge if ρ mediates the relation. Conditional tests and sparsity are required.

32.2 Forcing acyclicity

The physical system contains feedback. A DAG constraint applied to the static graph can delete or reverse valid edges. Use time unrolling or SCC condensation instead.

32.3 Using attention as proof of causality

Attention weights are model internals, not causal evidence. They must be validated through ablation and intervention.

32.4 Training on only one graph

If every sample uses the same adjacency, graph recovery is trivial memorization. Multiple mechanism combinations are required.

32.5 Common-source confounding

A single radiation pulse can move many variables together. Independent source interventions are needed to separate pathways.

32.6 Insufficient temporal resolution

Coarse sampling can make delayed edges appear instantaneous or reverse temporal precedence. Synthetic datasets should include multiple sampling rates.

32.7 Unobserved mediators

If ρ is omitted, the model may legitimately infer $C \rightarrow \phi$. Graph interpretation always depends on the chosen node set.

32.8 Physically impossible random samples

Independent randomization can violate Einstein relations, positivity, charge consistency, or material constraints. Use a constrained parameter generator.

32.9 Class imbalance

Absent edges dominate. Accuracy can appear high for a model that predicts no edges. Use precision-recall metrics and balanced losses.

32.10 Overly flexible candidate graph

Allowing every pair with no prior mask increases spurious shortcuts and data requirements. Begin with a physics-defined superset.

32.11 Synthetic-generator artifacts

The network may identify coding conventions or numerical noise instead of physics. Vary numerical tolerances and generator implementations in robustness tests, while never using solver order as a target.

33 Scientific claims that are and are not defensible

33.1 Defensible proof-of-concept claims

With successful synthetic tests, the project may claim that a physics-constrained GNN can recover known directed dependencies, edge types, lags, and feedback groups from trajectories and interventions within a specified model family. It may also claim that state-dependent edge strengths reveal when specific couplings become important.

33.2 Claims requiring experimental data

Synthetic success alone cannot establish that the graph is complete for a real irradiated silicon device, that inferred edge strengths are quantitatively calibrated to a particular fabrication process,

or that newly inferred edges correspond to real microscopic mechanisms. Those claims require independent experimental or higher-fidelity evidence.

34 Potential research contribution

A publishable contribution could combine:

1. a typed physical-variable graph spanning radiation damage, electrostatics, carrier transport, and thermal feedback;
2. graph-family synthetic generation with explicit mechanism switches;
3. intervention-supervised directed edge discovery;
4. separation of zero-lag algebraic relations from delayed dynamics;
5. SCC-based derivation of physical coupling groups and partial sequence;
6. physics-residual and conservation constraints;
7. uncertainty under hidden-mechanism discrepancy.

A precise title could be: *Physics-constrained graph neural discovery of directed multiphysics dependencies under radiation-induced material degradation.*

35 Extension to the superconducting branch

The same physical-graph framework can later represent nodes for quasiparticle density, pair-breaking phonons, trap/TLS occupation, offset charge, superconducting phase, junction current, parity, and temperature. Candidate edges would include pair-breaking phonon production, quasiparticle diffusion and recombination, trap-assisted poisoning, electrostatic offset-charge effects, and thermal feedback. The graph should remain a variable graph; device geometry remains global context or a source of node features.

36 Recommended development roadmap

1. Freeze the candidate node list and direct-edge definitions.
2. Implement explicit mechanism switches and continuous edge multipliers in the synthetic generator.
3. Build analytic chain, common-cause, and delayed-feedback tests.

4. Build the lumped Module 2–5 generator and generate multiple graph families.
5. Add matched interventions and graph labels.
6. Establish correlation, Granger-style, sparse-regression, MLP, fixed-graph, and NRI baselines.
7. Train the typed dependency MPNN with state and edge supervision.
8. Add physics residuals, conservation, sparsity, and prior losses.
9. Add lag prediction, SCC extraction, and condensation ordering.
10. Test noisy, missing-variable, and hidden-mechanism cases.
11. Move to field-simulation trajectories and POD or summary-node features.
12. Only after synthetic verification, seek experimental observables for partial graph validation.

37 Immediate first implementation

The first executable task should not be a large GNN. It should be a graph-labeled lumped generator with eight nodes:

$$\{C, T, \rho, \phi, E, n, p, J\}. \quad (37.1)$$

Implement five switchable mechanisms:

$$C \rightarrow \rho, \quad C \rightarrow \mu, \quad C \rightarrow \tau, \quad E \rightarrow C, \quad (J, E) \rightarrow T. \quad (37.2)$$

Generate 1,000–2,000 baseline cases and at least one matched intervention per active candidate edge. First verify that simple baselines and NRI can recover the graph. Then add typed messages and physics losses. This sequence isolates graph-learning difficulty before the full field model is introduced.

38 Summary

Module 11 now targets only the directed physical dependency graph connecting the variables and parameters of Modules 2–5. The graph nodes are physical fields, coefficients, rates, and external sources. Edges are direct mechanisms classified as algebraic, constraint, derivative, constitutive, transport, reaction, heating, or forcing relations.

Because the system contains feedback, the graph does not generally have one total ordering. A physically admissible coupling sequence is obtained by distinguishing zero-lag from delayed edges, identifying strongly connected feedback components, collapsing those components, and topologically ordering the condensation graph.

Synthetic data are sufficient for proof of concept when the generator varies not only parameters but also graph topology through explicit mechanism switches. Matched interventions are central: they distinguish direct edges from common-source correlations and quantify mechanism-specific effects. The recommended GNN is a directed typed message-passing encoder-decoder that predicts edge existence, type, strength, sign, lag, state trajectories, and uncertainty while obeying physics residuals and conservation constraints.

The most important limitation remains explicit: the learned graph describes the synthetic model family. Experimental or higher-fidelity data are required to claim completeness for a real irradiated device.

39 Appendix A: detailed candidate dependency table

Source	Target	Mechanism	Current status	sta-	Suggested intervention
C_j	ρ	charged-defect space charge	represented; reduced		set $z_j = 0$
n	ρ	electron charge	represented		clamp or remove electron variation
p	ρ	hole charge	represented		clamp or remove hole variation
N_D^+, N_A^-	ρ	fixed dopant charge	represented		vary dopant level
ρ	ϕ	Poisson constraint	represented		isolated charge forcing
ϕ	$\mathbf{E}$	spatial derivative	represented		analytic potential test
T	D_C	Arrhenius diffusivity	represented		isothermal clamp
D_C	C	defect diffusion	represented		scale D_C
$\mathbf{E}$	C_j	charged-defect drift	candidate; future	fu-	set defect mobility to zero
C_k	C_j	defect reactions	broader model		disable reaction channel
n, p	C_j	carrier-assisted reactions	candidate; future	fu-	remove capture terms
C	μ_n, μ_p	mobility degradation	represented; reduced		set degradation coefficient to zero
T	μ_n, μ_p	temperature-dependent mobility	represented; reduced		isothermal clamp
μ_n, T	D_n	Einstein relation	represented; reduced		independent consistency test
μ_p, T	D_p	Einstein relation	represented; reduced		independent consistency test
C	τ_n, τ_p	trap-assisted lifetime change	represented; reduced		set trap coupling to zero

Source	Target	Mechanism	Current status	Suggested intervention
T	τ_n, τ_p	thermally dependent capture	candidate	isothermal clamp
$\mathbf{E}, \mu_n, n, D_n$	$\mathbf{J}_n$	drift-diffusion current	represented	one-factor forcing tests
$\mathbf{E}, \mu_p, p, D_p$	$\mathbf{J}_p$	drift-diffusion current	represented	one-factor forcing tests
$\mathbf{J}_n, G, R$	n	continuity equation	represented	source or recombination ablation
$\mathbf{J}_p, G, R$	p	continuity equation	represented	source or recombination ablation
$\mathbf{J}, \mathbf{E}$	Q_J	Joule heat	represented; reduced	set $\gamma_J = 0$
Q_J	T	thermal source	represented; reduced	heat-source ablation
Q_{rad}	T	direct radiation heating	represented	heat-only excitation
R	T	recombination heating	broader model	set heat yield to zero
R_j	T	defect-reaction heat	broader model	disable reaction enthalpy

40 Appendix B: example graph-family table

Family	$C \rightarrow \rho$	$C \rightarrow \mu$	$C \rightarrow \tau$	$(J, E) \rightarrow T$
A: electrostatic damage	on	off	off	off
B: mobility damage	off	on	off	off
C: recombination damage	off	off	on	off
D: electrical self-heating	off	off	off	on
E: mixed damage	on	on	on	off
F: mixed electrothermal	on	on	on	on
G: null-coupling control	off	off	off	off

Family	$C \rightarrow \rho$	$C \rightarrow \mu$	$C \rightarrow \tau$	$(J, E) \rightarrow T$
H: continuous-strength	sampled γ	sampled γ	sampled γ	sampled γ

41 Appendix C: derivation of GCN normalization

Starting from adjacency $\mathbf{A}$, add self-loops:

$$\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}. \quad (41.1)$$

Define the degree matrix

$$\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}. \quad (41.2)$$

The symmetric normalized propagation operator is

$$\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2}. \quad (41.3)$$

Multiplying by this operator averages neighbor features with degree correction. For a directed typed physical graph, this standard operator is only a baseline; incoming and outgoing relations and edge types should usually be handled explicitly.

42 Appendix D: reproducibility checklist

1. Save all random seeds and generator versions.
2. Version the node list, candidate mask, and edge-type definitions.
3. Store raw parameters and derived coefficients.
4. Preserve baseline/intervention pairing identifiers.
5. Split by complete physical case before normalization.
6. Report all parameter and topology ranges.
7. Report failed synthetic cases and rejection reasons.
8. Save normalization statistics from training data only.
9. Evaluate at least five neural seeds.
10. Release reduced scripts for regenerating toy and lumped datasets.

11. Label every conceptual plot as illustrative.
12. Distinguish calibrated ranges from stress-test ranges.

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